

## FARMVILLE, VA, USA

WMO: 724017

Lat: 37.358N

Lon: 78.438W

Elev: 127

StdP: 99.81

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03707

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.2       | -7.3 | -20.0                         | 0.6 | -3.1 | -17.1 | 0.8 | -2.1 | 8.7                   | 9.7  | 7.9 | 9.1  | 0.7                   | 290  | 0.396 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.7                   | 35.2            | 24.2 | 33.7 | 24.1 | 32.4 | 23.6 | 26.9                | 31.9 | 25.9 | 30.8 | 25.2 | 30.1 | 2.6                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 25.9                            | 21.6 | 29.4 | 24.2 | 19.4 | 27.9 | 23.8 | 19.0 | 27.5 | 84.9          | 31.8 | 80.6 | 31.4 | 77.7 | 30.0 | 30.5           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.4           | 6.0  | 5.2 | DB                         | -15.0 | 37.1               | 3.9 | 2.2                                                | -17.8 | 38.7       | -20.1 | 40.0       | -22.3 | 41.2       | -25.1 | 42.8 | (4) |
| (5)               |      |     | WB                         | -15.6 | 27.6               | 3.8 | 1.3                                                | -18.3 | 28.5       | -20.5 | 29.3       | -22.6 | 30.0       | -25.4 | 31.0 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 14.7   | 3.5   | 4.8   | 9.3   | 14.7  | 19.1  | 23.5  | 25.8  | 25.0  | 21.1  | 15.0  | 9.0   | 5.4   | (6)  |
| (7)  |                                                                                  | DBStd       | 9.05   | 5.86  | 5.40  | 5.48  | 4.88  | 4.16  | 3.28  | 2.60  | 2.69  | 3.61  | 4.52  | 5.08  | 5.07  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 717    | 216   | 160   | 82    | 11    | 0     | 0     | 0     | 0     | 0     | 8     | 79    | 161   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2124   | 459   | 379   | 285   | 131   | 44    | 3     | 0     | 0     | 16    | 124   | 282   | 402   | (9)  |
| (10) |                                                                                  | CDD10.0     | 2446   | 14    | 14    | 59    | 151   | 281   | 406   | 490   | 466   | 334   | 162   | 49    | 18    | (10) |
| (11) |                                                                                  | CDD18.3     | 812    | 1     | 0     | 4     | 21    | 67    | 159   | 232   | 208   | 99    | 20    | 2     | 1     | (11) |
| (12) |                                                                                  | CDH23.3     | 8149   | 2     | 3     | 51    | 278   | 677   | 1590  | 2374  | 2043  | 889   | 218   | 19    | 4     | (12) |
| (13) | CDH26.7                                                                          | 3407        | 0      | 1     | 7     | 77    | 216   | 668   | 1127  | 925   | 327   | 59    | 1     | 0     | (13)  |      |
| (14) | Wind                                                                             | WSAvg       | 1.7    | 2.0   | 2.0   | 2.2   | 2.3   | 1.8   | 1.5   | 1.4   | 1.3   | 1.5   | 1.5   | 1.7   | 1.7   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1098   | 90    | 83    | 103   | 77    | 103   | 87    | 111   | 94    | 88    | 82    | 83    | (15)  |      |
| (16) |                                                                                  | PrecMax     | 1468   | 207   | 171   | 239   | 183   | 222   | 230   | 202   | 287   | 273   | 309   | 243   | 203   | (16) |
| (17) |                                                                                  | PrecMin     | 803    | 23    | 16    | 10    | 5     | 35    | 13    | 29    | 9     | 2     | 0     | 9     | 8     | (17) |
| (18) |                                                                                  | PrecStd     | 174    | 43    | 40    | 49    | 38    | 44    | 56    | 52    | 59    | 66    | 66    | 50    | 45    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 22.1  | 22.8  | 27.6  | 32.0  | 32.6  | 36.2  | 37.8  | 37.1  | 34.1  | 31.2  | 26.1  | 22.8  | (19) |
| (20) |                                                                                  |             | MCWB   | 16.3  | 14.9  | 16.5  | 19.4  | 22.8  | 24.3  | 25.0  | 25.4  | 22.3  | 21.5  | 17.6  | 16.4  | (20) |
| (21) |                                                                                  | 2%          | DB     | 18.8  | 19.8  | 24.6  | 28.8  | 31.1  | 33.7  | 36.0  | 34.7  | 32.4  | 27.9  | 22.8  | 19.1  | (21) |
| (22) |                                                                                  |             | MCWB   | 14.9  | 12.8  | 15.4  | 18.3  | 21.9  | 24.5  | 24.5  | 24.7  | 22.4  | 19.9  | 16.4  | 16.0  | (22) |
| (23) |                                                                                  | 5%          | DB     | 16.4  | 17.2  | 22.2  | 27.1  | 29.0  | 32.3  | 33.8  | 32.9  | 30.5  | 26.0  | 21.0  | 17.2  | (23) |
| (24) |                                                                                  |             | MCWB   | 11.9  | 11.2  | 13.8  | 17.2  | 21.1  | 23.8  | 24.3  | 24.1  | 22.0  | 19.1  | 15.6  | 13.2  | (24) |
| (25) |                                                                                  | 10%         | DB     | 13.0  | 14.0  | 19.1  | 24.0  | 27.4  | 30.9  | 32.4  | 31.8  | 28.0  | 23.0  | 18.1  | 14.1  | (25) |
| (26) |                                                                                  |             | MCWB   | 9.0   | 9.0   | 12.6  | 16.1  | 20.2  | 23.1  | 24.1  | 23.6  | 21.2  | 17.5  | 13.8  | 10.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 17.9  | 17.0  | 18.8  | 21.9  | 24.6  | 27.2  | 27.9  | 28.1  | 25.8  | 24.1  | 20.5  | 19.1  | (27) |
| (28) |                                                                                  |             | MCD B  | 19.9  | 20.7  | 24.2  | 27.7  | 30.0  | 32.9  | 33.9  | 32.6  | 29.5  | 26.5  | 22.4  | 20.9  | (28) |
| (29) |                                                                                  | 2%          | WB     | 15.7  | 15.0  | 17.2  | 20.2  | 23.3  | 25.8  | 26.9  | 26.8  | 24.5  | 22.2  | 18.7  | 16.8  | (29) |
| (30) |                                                                                  |             | MCD B  | 18.3  | 18.2  | 21.9  | 25.8  | 28.5  | 30.9  | 32.2  | 31.5  | 28.0  | 26.1  | 21.1  | 18.7  | (30) |
| (31) |                                                                                  | 5%          | WB     | 12.9  | 12.1  | 15.5  | 18.9  | 22.3  | 25.0  | 25.9  | 25.8  | 23.5  | 20.5  | 17.0  | 14.0  | (31) |
| (32) |                                                                                  |             | MCD B  | 15.6  | 15.8  | 20.4  | 23.5  | 27.3  | 29.9  | 30.8  | 30.8  | 27.3  | 24.0  | 19.6  | 16.6  | (32) |
| (33) |                                                                                  | 10%         | WB     | 9.6   | 9.4   | 13.7  | 17.5  | 21.2  | 24.1  | 25.2  | 25.0  | 22.7  | 18.8  | 14.8  | 11.0  | (33) |
| (34) |                                                                                  |             | MCD B  | 12.7  | 13.6  | 18.2  | 22.0  | 25.7  | 28.8  | 30.2  | 29.8  | 26.6  | 22.3  | 17.9  | 13.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.5  | 12.5  | 13.4  | 13.9  | 12.7  | 12.0  | 11.7  | 11.9  | 12.2  | 13.5  | 13.4  | 11.7  | (35) |
| (36) |                                                                                  |             | MCDBR  | 14.4  | 16.4  | 16.3  | 16.7  | 14.5  | 13.7  | 13.9  | 13.7  | 14.8  | 15.7  | 16.1  | 14.5  | (36) |
| (37) |                                                                                  |             | MCWBR  | 10.4  | 10.8  | 9.0   | 7.9   | 6.9   | 5.8   | 5.2   | 5.3   | 6.2   | 8.4   | 10.1  | 10.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 12.6  | 14.4  | 13.7  | 13.3  | 12.4  | 11.9  | 11.5  | 11.4  | 11.2  | 12.4  | 12.0  | 12.2  | (38) |
| (39) |                                                                                  |             | MCWBR  | 10.6  | 11.2  | 8.9   | 7.6   | 6.5   | 5.7   | 5.2   | 5.2   | 5.7   | 7.6   | 9.2   | 10.1  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.298 | 0.313 | 0.347 | 0.384 | 0.430 | 0.477 | 0.508 | 0.506 | 0.403 | 0.358 | 0.319 | 0.302 | (40) |
| (41) |                                                                                  | taud        |        | 2.526 | 2.474 | 2.421 | 2.334 | 2.222 | 2.139 | 2.075 | 2.062 | 2.341 | 2.456 | 2.525 | 2.552 | (41) |
| (42) |                                                                                  | Ebn at Noon |        | 898   | 923   | 917   | 896   | 856   | 813   | 784   | 775   | 846   | 858   | 863   | 869   | (42) |
| (43) |                                                                                  | Edh at Noon |        | 78    | 92    | 107   | 124   | 141   | 153   | 163   | 161   | 115   | 94    | 78    | 71    | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 2.38  | 3.13  | 4.15  | 5.15  | 5.78  | 6.31  | 5.96  | 5.38  | 4.43  | 3.53  | 2.64  | 2.07  | (44) |
| (45) |                                                                                  | RadStd      |        | 0.17  | 0.37  | 0.28  | 0.40  | 0.49  | 0.42  | 0.34  | 0.27  | 0.45  | 0.41  | 0.19  | 0.18  | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) | Station Only            | +0.71 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |  |  |
| (47) | Regional (54 neighbors) | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +147    | +93     |  |  |

Nomenclature: See separate page